

# NXSG State, research & validation

A complete product and engineering writeup of NX Shader Graph 1.5.0.

NXSG is a visual shader editor for Unity Built-In, developed for PC VRChat avatars. The current package combines a substantial node library with generated shaders, material tools and a portable graph format.

## 155 addable nodes

Eight library categories. Counted from the current catalog and Add menu filter.

## 35 example graphs

293 authored node instances and 297 connections across the bundled samples.

## Stable 1.5.0 release

The live VPM ZIP matches its published SHA-256 in this report's verification.

## Validation still has limits

Linux editor/render evidence and D3D11 cross-compilation; native Windows and headset results remain open.

## WHERE THE PROJECT SHOULD GO NEXT

Consolidate the feature set: close client/stereo gaps, measure expensive effects on representative hardware, make the documentation match the code, and only then extend the portable format into new tools.

This report covers current features, future options, measured timings, recorded regressions, research decisions, architecture, rendering techniques, integrations, the complete node inventory and every bundled example. It audits existing evidence; it is not a claim that a new full regression suite was run today.

Prepared 25 September 2026 | Source revision 4e36698 | Evidence, diagrams and calculations are labeled throughout.

# Where we are now

A released authoring tool with broad functionality and an uneven evidence boundary.

| Area                | Current position                                                                                                                           |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Package             | NX Shader Graph 1.5.0 / dev.nerdrx.nxsg. Available through ALCOM and Creator Companion without prerelease mode.                            |
| Target              | Unity 2022.3.22f1; Built-In forward rendering; PC VRChat. The SDK fixture uses Base/Avatars 3.10.5.                                        |
| Development fixture | Linux, OpenGLCore and an RX 7900 XTX. Hidden Gamescope is used for isolated editor/render checks.                                          |
| Graph ownership     | .nxsg is editable source. ShaderLab/HLSL and Unity materials are generated outputs.                                                        |
| Latest release      | Layered PBR clearcoat and colored sheen, Cost estimates, resource cleanup and validation/build/package hardening.                          |
| Not established     | Native Windows rendering, live client/mirror/stereo equivalence and representative headset GPU performance. Quest avatars are unsupported. |

## How to read the rest

**Current state and future:** pages 1-5. **Testing and measurement:** pages 6-14. **Research, design and feature behavior:** pages 15-31. **Full catalogs and sources:** pages 32-39.

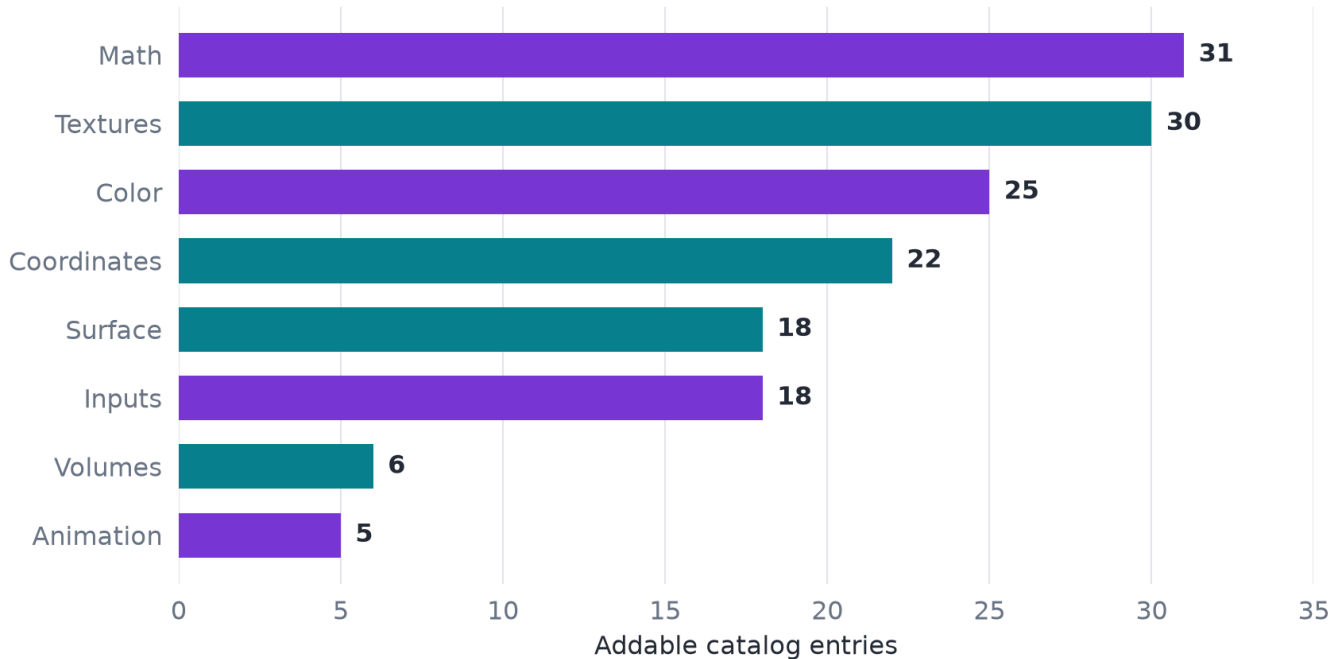
### PUBLIC SOURCE IS NOT A LICENSE GRANT

The repository is public, but the current README says no project-wide open-source license has been selected. Third-party material retains its own notices and licenses.

Sources: R01 README; R02 validation; R03 inventory; R04 compatibility; W01 Unity pin.

# The actual feature footprint

Catalog and sample counts were derived directly from the current code and assets.



**Fig. 1. INVENTORY.** 155 Add-menu entries at revision 4e36698; the hidden preview helper is excluded. Category membership follows the editor.

The registry has 156 operations in total: 104 base catalog operations and 52 feature entries. One internal operation, `core.previewVector`, is hidden from the Add menu. These are implementation counts, not a measure of creative value or test coverage.

The feature families include Toon/PBR/Unlit, clearcoat and sheen, procedural textures, UV transforms, color grading, fur, shells, particles, tessellation, parallax, volumes, AudioLink and optional LTCGI. Editing includes live previews, Patterns, texture naming, copy/paste, property exposure, presets and build recovery.

## DOCUMENTATION CORRECTION APPLIED

The audited revision of docs/NODES.md said 143 visible nodes and 45 feature nodes. Local documentation now says 155 and 52, and its feature-name list matches the catalog. These documentation fixes have not been published.

# The next gains are confidence and control

Proposed priorities, not an approved schedule or shipped features.

| Priority             | Work                                                                                                            | Evidence that would close it                                                                                  |
|----------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| 1 / Verify           | Native Windows, VRChat client, mirrors and headset stereo. Reproduce the reported avatar cases in known worlds. | Named fixture, exact versions, paired per-eye images, logs and a reproducible graph.                          |
| 2 / Measure          | GPU cost of shells/fins, particles, volumes, pixel lights and transparency on target hardware.                  | Matched scenes and controls; frame-time distributions, overdraw/camera context, no invented universal budget. |
| 3 / Consolidate      | Synchronize docs with catalog; improve regression fixtures, accessibility and display-scale checks.             | Generated inventory, schema checks, dense-graph interaction tests and minimum-window/DPI captures.            |
| 4 / Simplify use     | Performance-oriented presets and explanations tied to measured effect families.                                 | Profiles justified by measured hardware, with explicit quality tradeoffs and editable limits.                 |
| 5 / Extend carefully | Node SDK, portable tooling, Blender bridge, CLI and viewer.                                                     | Versioned contracts and explicit exact/approximate/baked/unsupported mappings.                                |

A release number signals package stability and availability. It cannot substitute for device evidence. The strongest next milestone is a compact, reproducible avatar validation set that can be rerun after compiler or render-state changes.

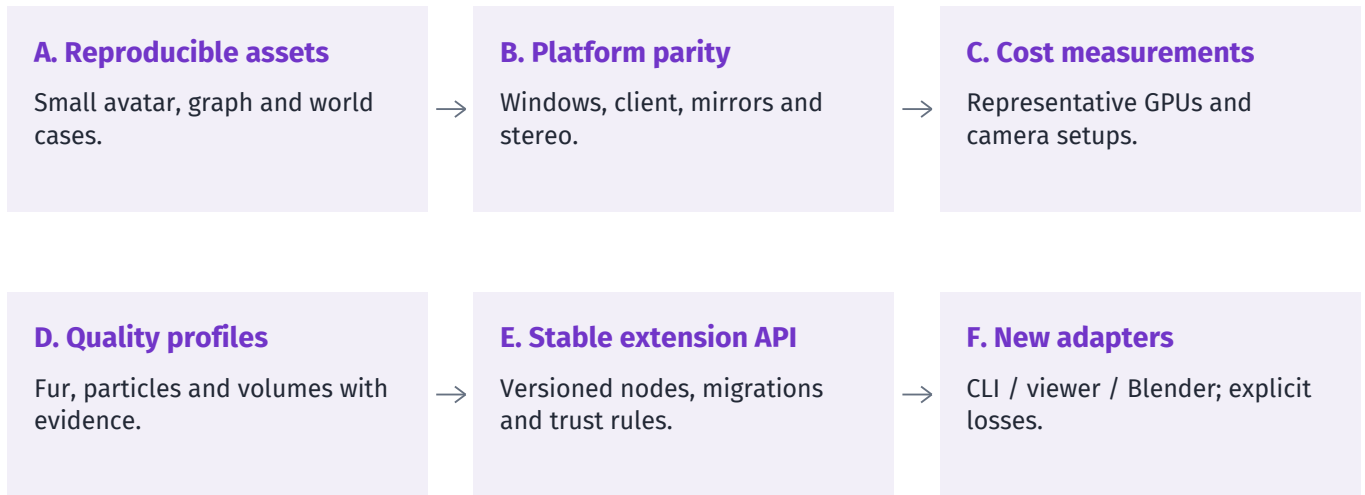
## KEEP THE CREATIVE SCOPE

There is already enough material variety to build interesting work. Further nodes should fill a demonstrated gap or improve an existing workflow, while the validation base catches up.

Sources: R01 priorities; R04 compatibility; R06 risk register; R07 implementation gates. Recommendations are report synthesis.

# A roadmap with explicit gates

Dependencies make more sense here than promised dates.



**Fig. 2.** PROPOSAL. A dependency sketch; rows show related work, not guaranteed delivery order.

## What is possible, but not automatic

- **Portable tools:** the core separation makes them plausible. It does not make Unity shader semantics transferable without an adapter.
- **Mobile:** a future path would map or bake to VRChat-permitted materials. Arbitrary custom avatar shaders are not the current Quest path.
- **Self-shadowing:** local fur approximations can improve a look. Cross-body occlusion needs appropriate scene information; unrelated material slots do not acquire shared geometry knowledge through a normal material pass.
- **Performance presets:** useful once measured. A node count or a static pass estimate alone cannot identify a safe headset preset.

### DECISION RULE

Prefer a new capability when its use case, compatibility boundary and validation fixture can be stated together. Preserve current graphs and stable material properties as the project grows.

# What the evidence actually proves

A passed check establishes its own scope. It does not automatically establish the next environment.

|             |                              |          |
|-------------|------------------------------|----------|
| <div></div> | Portable contracts           | RECORDED |
| <div></div> | Linux editor interaction     | RECORDED |
| <div></div> | Linux rendered output        | RECORDED |
| <div></div> | Windows-target D3D11 compile | RECORDED |
| <div></div> | Native Windows render        | OPEN     |
| <div></div> | Live AudioLink / VRCFury     | OPEN     |
| <div></div> | VRChat world + mirrors       | OPEN     |
| <div></div> | Stereo headset + GPU timing  | OPEN     |

**Fig. 3.** EVIDENCE MAP. Green means recorded evidence in the project; amber means a remaining acceptance area. This is not a coverage percentage.

Portable checks exercise graph meaning without Unity. Editor checks exercise the real pinned Unity UI, often through synthetic events. Render checks compare controlled pixels or geometry. D3D11 bundle builds cross-compile selected shaders on Linux. Native Windows and live client/headset results require their own runs.

| Evidence label used here | Meaning                                                                       |
|--------------------------|-------------------------------------------------------------------------------|
| MEASURED                 | A recorded timing or other observation, with its environment and sample size. |
| INVENTORY                | A count derived from current source/assets, not runtime quality.              |
| MODEL / EXPLANATION      | An equation or conceptual diagram, not an observed GPU result.                |
| PROPOSAL                 | Work or a validation method recommended for the future.                       |

Sources: R02 validation; R04 compatibility; R10 retained v1.5 logs. Historical evidence, audited today.

# The 1.5 quality pass

The latest retained logs match the principal success markers named in the validation record.

| Area               | Exercised behavior                                                                                                                                                                                 |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Portable core      | Layer sockets, save/load, neutral PBR parity, dynamic zero weights, disabled dependency pruning and invalid properties. Static pass estimates checked against every shipped sample's declarations. |
| Layered rendering  | Finite pixels; independent coat roughness/normal; sheen roughness and black tint; emission, opacity and additional lights.                                                                         |
| Layered controls   | Header switching, connection retention, field identity, connected controls and inspector bounds.                                                                                                   |
| Cost panel         | 240, 310 and 400 px sidebar widths; texture names; clicking a hotspot focuses its node and opens the Inspector.                                                                                    |
| Resource lifetime  | Deleting nodes repeatedly while preview is paused destroys associated thumbnail editors, materials and shaders.                                                                                    |
| Build / scene      | Changed and unchanged builds, stable GUIDs, corrupt or missing outputs, compiler-failure rollback, scene updates and Auto scene off.                                                               |
| Target compilation | 16 selected shaders cross-compiled to a Windows-target D3D11 asset bundle.                                                                                                                         |
| Packaging          | 11 documented tests, including same-version ZIP conflict preservation and rejection of untracked package files.                                                                                    |

The thumbnail test checks ownership and disposal of named resources. It does not establish that Unity's total resident memory remains flat over arbitrary sessions. The Cost panel tests layout and static estimates; they do not time the GPU.

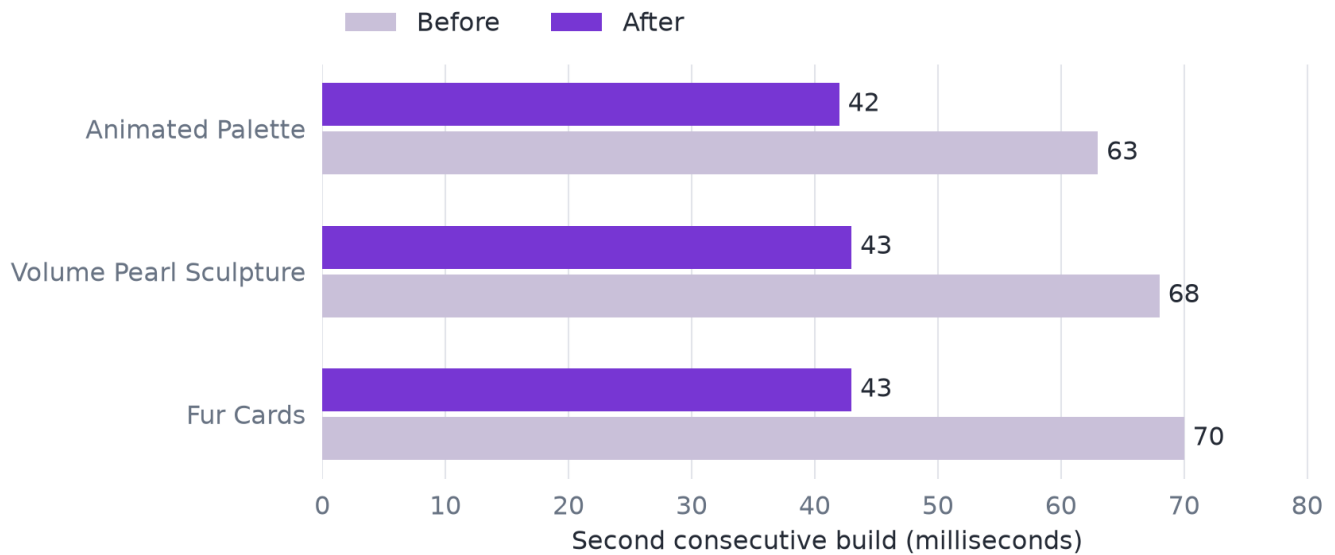
### IMPORTANT CONTINUITY RULE

A disconnected layer whose fixed weight is zero may be omitted. A connected or animated weight that currently happens to be zero must remain live.

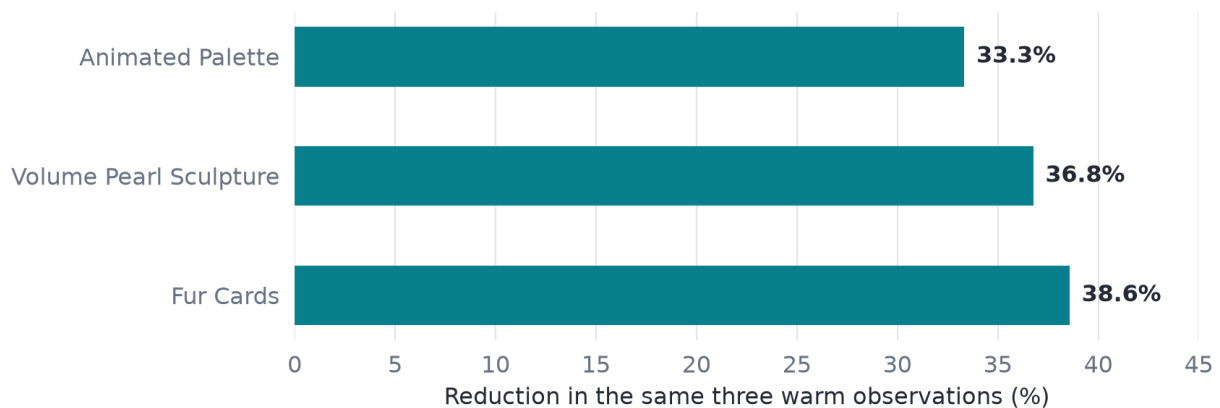
Sources: R02:568-580; R10 log manifest on page 39. Package-test total is documented in R02.

# Build for VRChat: measured improvement

Three second-consecutive-build observations from the same Linux/OpenGL fixture, 22 September 2026.



**Fig. 4.** MEASURED. One warm observation per graph before and after. Existing Unity shader caches; no error bars or repeated-run distribution were recorded.



**Fig. 5.** DERIVED FROM THE SAME OBSERVATIONS. Reductions are 33.3%, 36.8% and 38.6%; these are not additional measurements.

The removed staging import eliminates a duplicate shader import/validation stage. Unchanged emitted text is no longer rewritten or force-imported. Generation still runs, dependencies are checked and material bindings still apply. Save imports the graph rather than refreshing the whole project.

**Limits:** these timings exclude the project-wide-refresh improvement and are not cold-cache, whole-avatar, Windows or SDK-upload benchmarks. They do not predict your project's full build duration.

Sources: R11 BUILD\_PERFORMANCE.md:11-21. Percentages calculated from the documented milliseconds.

# Responsiveness without invented FPS

One measured editor path and several behavior checks tell different stories.

## 150 nodes / 148 wires

The v1.4 synthetic editor fixture size.

## 1,000 updates / 138.6 ms

Recorded aggregate for warm insertion-target updates.

Dividing the recorded aggregate gives **0.1386 ms per update on average**. This is a derived value for that path only. It is not frame time, event-to-display latency or a before/after comparison. The raw original timing log was not matched during this audit; the number is retained in the validation document.

### Graph revision

Invalidates relevant cached socket geometry.



### Pointer operation

Uses the rebuilt cache for insertion targets.



### Scene synchronization

Defers checks; skips layout-only changes.

**Fig. 6.** EXPLANATION. The tested behavior targets specific editor work, not a universal speed claim.

## What was checked beyond speed

- Numeric-field identity is retained during edits; nonfinite input is rejected; reset and Undo are exercised.
- A 150-node fixture checks cache rebuilding and wire insertion/Undo. Auto scene off and last-valid-shader behavior remain respected.
- The UI pass checks wide and 850 x 500 layouts, a collapsible preview, numeric bounds and the connected-node picker near the canvas edge.
- The latest cleanup checks release thumbnail objects after deletion even while preview is paused.

### NEXT MEASUREMENT

Record p50/p95 interaction latency over repeated runs at 20, 200 and larger graph sizes, including reload, search, dragging, wire insertion and undo. Keep editor CPU and rendering cost separate.

# Interaction and material correctness

These are real fixture assertions, not a claim of exhaustive usability testing.

| Family                    | Recorded checks and useful boundaries                                                                                                                             |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sockets / selection       | Connections in both directions; compatible-node menu; mismatch rejection; Escape; outlines; box select; group drag/delete and Undo. Many use synthetic UI events. |
| Clipboard                 | Fresh IDs, internal wires, resource/parameter remapping, target unchanged on failure, shortcut isolation and copy/duplicate behavior.                             |
| Operation switching       | Node identity/layout/selection survive; compatible connections persist; inappropriate connections are removed; one Undo restores the change.                      |
| Preview lifecycle         | Unsaved graphs render; source material stays intact; invalid edits retain the previous valid preview; resources are disposed on close.                            |
| Color / numeric rendering | Red constant regression; scalar/color inference; signed safe divide; ramps; Gamma/Linear hue-space comparisons; color-mask/replacement controls.                  |
| Material identity         | Graph/material/shader context shown; readable generated names; stable GUIDs and overrides retained across successful builds.                                      |
| Curve editing             | Particle curves survive data roundtrip and editing; controls represent the authored curve rather than stale identity values.                                      |

## What remains a human review task

Keyboard access, high-DPI/theme combinations, screen-reader support, discovery by new users and long-session behavior need deliberate coverage. The Linux keyboard-Undo issue remains documented; visible toolbar controls provide an alternative.

### USABILITY FINDINGS SHOULD BE REPRODUCIBLE

Capture the exact window size, display scale, graph, focused field and action. A screenshot can show alignment, but cannot establish keyboard behavior or save/rebuild correctness.

Sources: R02 dated editor/render sections; R12 UI\_POLISH; R13 CREATOR\_WORKFLOW.

# Three bugs worth remembering

The visible symptom was misleading in each case.

| Reported symptom                             | Established cause / change                                                                                                                                                                                             | Remaining boundary                                                                                            |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Opaque material appears faintly transparent  | Camera-dependent albedo was incorrectly treated as coverage when Use albedo alpha was off. This could remove ShadowCaster and its depth contribution. Dependency analysis now follows albedo only when alpha is used.  | Controlled depth test passes. The particular VRChat world still needs a client retest.                        |
| Particles absent on hair only                | The privately inspected hair region had 4,118 vertices with RGB 0 and alpha 0. Implicit source-vertex tint/alpha suppressed particles. Particle lifetime alpha is now separate; vertex color is opt-in through wiring. | Same-pixel comparison for black/white source vertices passes locally. This is not a live client confirmation. |
| Seamless panorama shows seams, then speckles | Wrapped U derivatives caused filtering discontinuities. Alternate charts and stable derivative selection address the seam; a tolerance avoids neighboring pixels choosing charts differently.                          | Fractional tiling has a GPU regression. Headset panorama behavior remains a separate gate.                    |

A material can have Opacity = 1 while still interacting incorrectly with scene depth. A disconnected mask can leave an implicit vertex-alpha multiplier active. A seamless image can still be sampled with discontinuous derivatives. Diagnose the generated path and hidden inputs before changing the artwork.

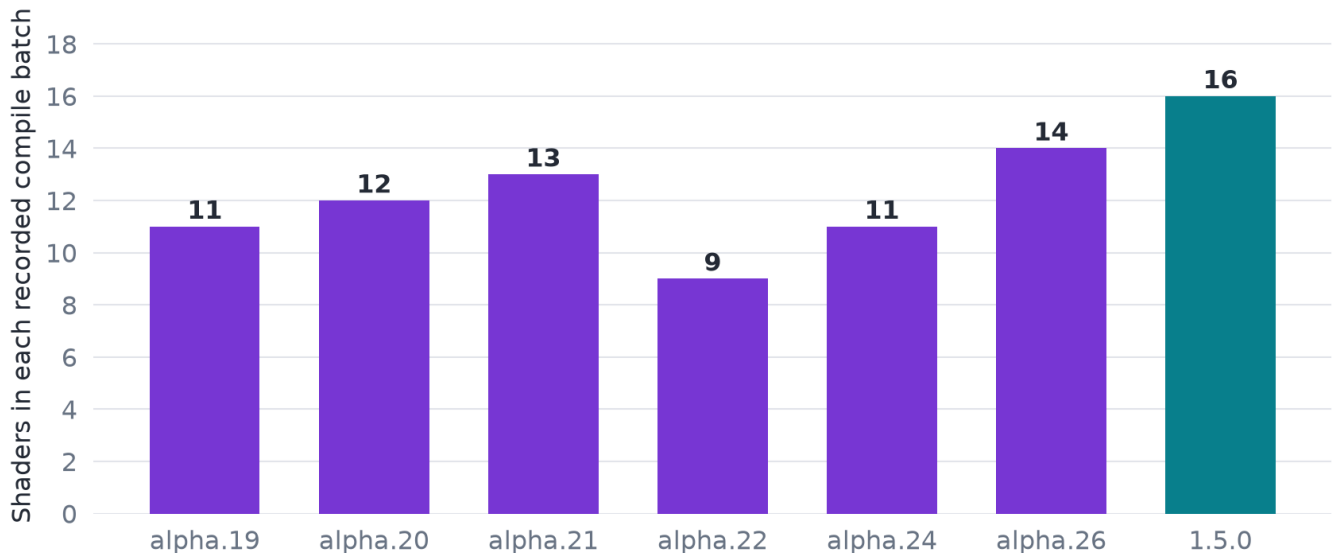
## REGRESSION STRATEGY

Keep a minimal graph and a controlled render case for each failure. Preserve the original cause in the test name or assertion so later optimizations do not reintroduce it.

Sources: R02 alpha.19-21 entries; R14 PANOSPHERE; W02 Unity depth textures. Private avatar assets are not reproduced.

# D3D11: stronger evidence, specific scope

The selected cross-compilation batch grew, but its membership changed between runs.



**Fig. 7. RECORDED BATCH INVENTORY.** Counts are separate test batches, not cumulative passes, percentage coverage or a smooth quality trend.

The reported X4571 failure was a vertex-output budget issue in the advanced path. Shared input data plus lighting/shadow/stereo outputs could exceed the SM4.0 budget. Advanced generated passes now require interpolators32; the compact basic Toon route retains its smaller path.

The latest batch contains 16 selected shaders, including layered-surface examples, tessellation, fur cards, particles, volumes and an unoptimized PBR variant. Successful Windows-target bundle compilation catches compiler/variant problems that an OpenGL render alone may miss.

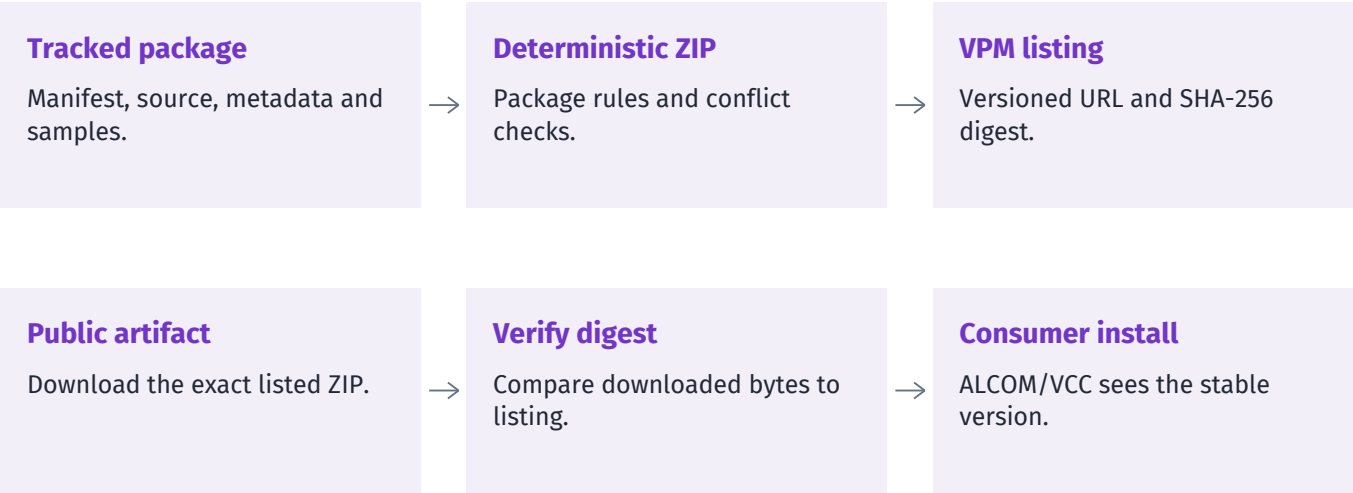
## WHAT IT DOES NOT SHOW

The compiler ran under Linux Unity. It did not render those shaders in a native Windows editor or VRChat client, test both headset eyes, exercise every light/shadow keyword, or establish GPU performance.

The alpha.22 count drops to nine because the fixture selection differs. Treating this chart as a regression in quality would be incorrect.

# Release integrity and repeatable builds

The live 1.5.0 download was checked again while preparing this report.



**Fig. 8.** RELEASE PROCESS. This report rechecked listing-to-download integrity; it did not reinstall the package into an avatar project.

| Current verification             | Result                                                           |
|----------------------------------|------------------------------------------------------------------|
| Version                          | 1.5.0                                                            |
| ZIP size                         | 276,502 bytes                                                    |
| Published and downloaded SHA-256 | 4523818bddb03b68f8eac03f2db7e8c9f340747dd6465d3921110d8491ed1d2d |
| Digest match                     | True                                                             |
| Repository snapshot              | 4e3669888d1f                                                     |

Stable shader/material GUIDs protect scene references through rebuilding. The transactional build journals existing output before mutation and restores assets at tested failure checkpoints. A rejected conflicting same-version package build must leave the existing release ZIP intact.

## TWO DIFFERENT ASSURANCES

Checksum equality establishes artifact integrity. Successful install, shader rendering and client acceptance are separate checks. Packaging tests are useful because they protect what users receive; they cannot validate a headset image.

Sources: R15 VPM scripts and listing; source/evidence.json live snapshot; R02 packaging checks; W03 VPM format.

# The remaining validation program

A concrete way to replace unknowns with reusable evidence.

| Fixture             | Vary                                                                                     | Record                                                                                    |
|---------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Client avatar       | Small baseline plus body/hair material slots; simple and layered surfaces.               | Exact world/client/SDK, graph hash, material state and comparison views.                  |
| Mirrors / stereo    | Direct view, mirror, each eye; opaque and transparent backgrounds.                       | Paired images and repro steps for alpha/depth/culling artifacts.                          |
| Expensive effects   | Shell count, local shadow samples, particle rate/lifetime, volume steps and light count. | GPU frame-time distributions with a plain-material control; coverage and camera counts.   |
| Integrations        | AudioLink absent/present/no data; VRCFury property animation; LTCGI sources on/off.      | Expected value/color response, fallback behavior, versions and changed property IDs.      |
| Editor lifecycle    | Large graphs, reload/close, repeated node deletion, undo, invalid edits.                 | Latency distribution and owned-object counts; total process memory as a separate measure. |
| Hardware / platform | Native Windows plus more than the existing AMD Linux fixture.                            | Driver/API versions, shader variant list and any visual divergence.                       |

Use repeatable camera paths and fixed graph/time inputs where possible. Warm up separately, report distributions rather than one best run, and retain raw results. Record missing data explicitly. A small well-described benchmark suite is more useful than a large unlabeled “works” gallery.

## NO PROJECT-WIDE PASS PERCENTAGE

The repository contains 72 Editor C# files and 40 Portable C# files. Those are file inventories, not counts of test cases or proof that every fixture ran at this revision.

Sources: R04 compatibility; R06 risks; R07 implementation plan; report recommendations.

# What was researched

The initial research baseline is dated 17 September; later implementation evidence supersedes its early pending status.

| Research strand      | Resulting direction                                                                                                     | Open question                                                            |
|----------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Graph / compiler     | Portable JSON, stable identities, typed DAG, deterministic emission and conservative specialization.                    | How much portability can adapters preserve without silent semantic loss? |
| Editor / prior art   | Familiar sockets, search, previews and Undo; replaceable canvas adapter.                                                | Accessibility and large-graph interaction across OS/DPI combinations.    |
| Platform / AudioLink | Pinned PC target, stable material properties, optional texture-data provider and missing-data fallback.                 | Live client, headset, missing-provider and animation behavior.           |
| Fur / rendering      | Treat geometry generation, overdraw and camera work as separate cost drivers.                                           | Measured quality budgets for representative avatars and devices.         |
| Motion / particles   | GPU feasibility research compared material, extra-mesh and same-mesh paths. Analytical effects fit the shipped request. | Pose-following versus stored world-space history must stay explicit.     |
| Release / trust      | Bounded input handling, exact-file license review, deterministic package verification.                                  | Public license policy and safe extension distribution.                   |

The index records 123 distinct linked source URLs as reachable on 17 September. Link availability is a documentation check, not evidence that the linked claims or NXSG behavior were validated. This report re-opened selected primary sources for the platform, depth, color and packaging points.

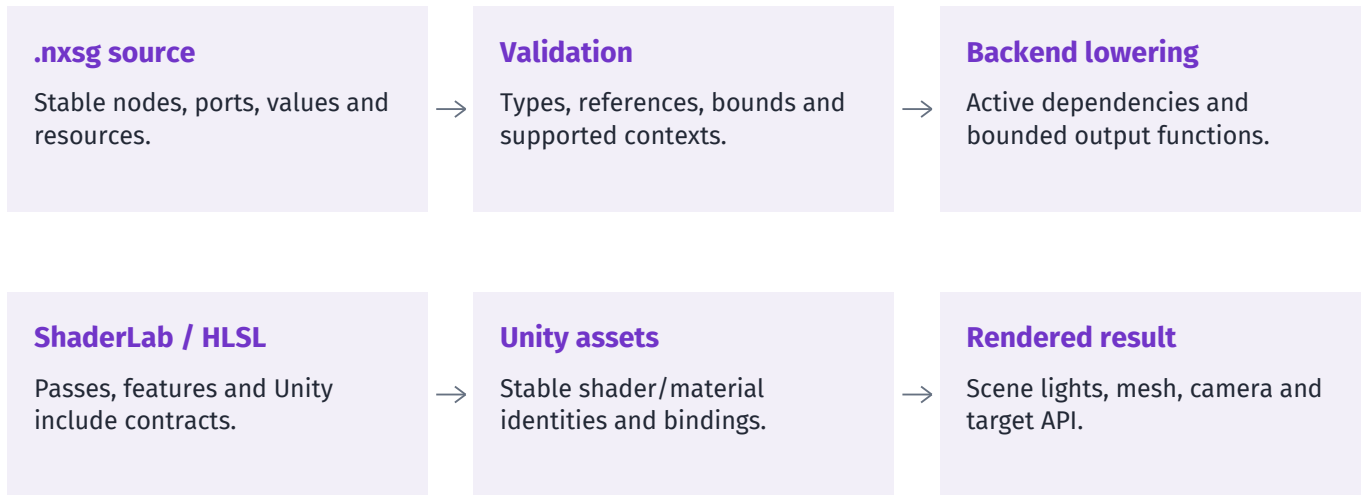
## RESEARCH AND PRODUCT ARE DIFFERENT SNAPSHOTS

Early reports say fur, Patterns and baking are planned. The current package contains implementations of those families. Read dated research as the rationale and current source/validation as the implementation record.

Sources: R16 research index; R08 decisions; R09 compiler research; R17 rendering; R18 platform; R19 UX; R20 release research.

# A portable graph, a concrete backend

The file format owns author intent; Unity owns editor and generated-asset integration.



**Fig. 9.** IMPLEMENTED FLOW, simplified. Layout and Unity adapters sit around the graph; future tools must preserve its meaning.

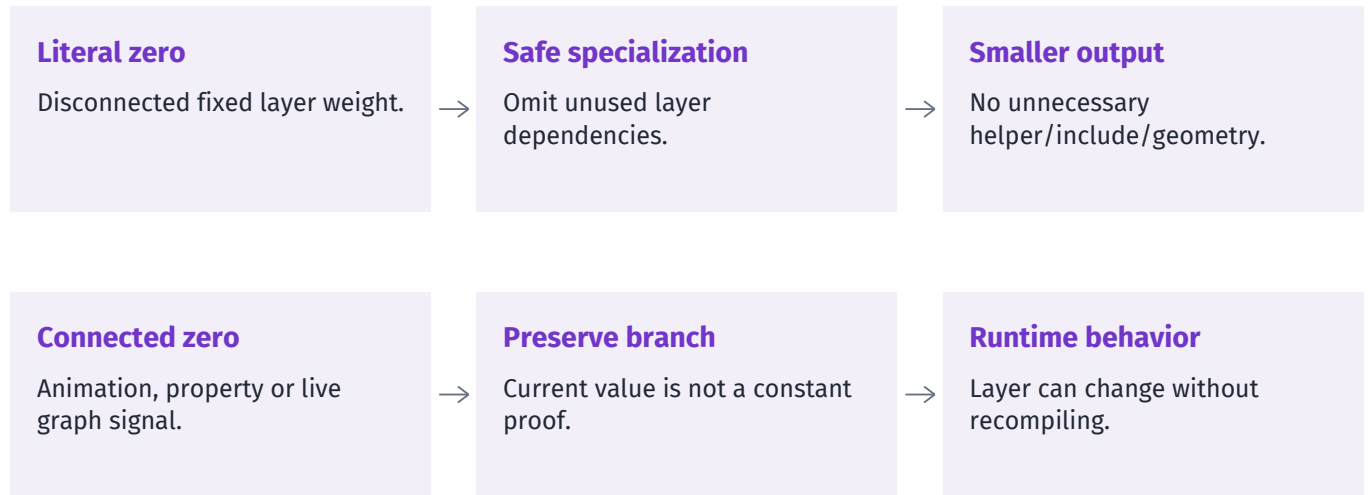
## Why this boundary matters

- The graph can be moved or serialized without making a canvas implementation the source of truth.
- Generated shaders can be rebuilt after a compiler fix; stable asset identities keep material and scene references intact.
- Unknown operations are retained as inert data with diagnostics. Opening an unfamiliar graph is not permission to execute a plugin.
- Preview and final build must share graph semantics even when their asset lifetimes and rendering contexts differ.
- A future CLI or viewer can inspect the same graph, but it cannot establish Unity compilation or VRChat acceptance by itself.

The original design proposed a richer typed IR and several future adapters. This report does not claim that every proposed optimizer or backend layer is implemented. Current lowering and validation are the practical contract.

# Types, identity and mutability

A zero value can mean “off forever” or “off right now.” The compiler must distinguish them.



**Fig. 10.** SEMANTIC RULE. The second row is deliberately retained even when its current numeric value equals the first row.

Connections name stable node and port IDs. Parameters and resources have their own stable IDs. Clipboard operations remap identities while retaining internal relationships. Header operation switching preserves the node identity and only keeps compatible wires.

Numeric operations infer scalar or color behavior from graph demand. Scalar promotion and color-to-number use are explicit compiler conversions. Color-to-opacity wiring must have defined grayscale semantics; alpha is separately available through Texture Alpha or split channels.

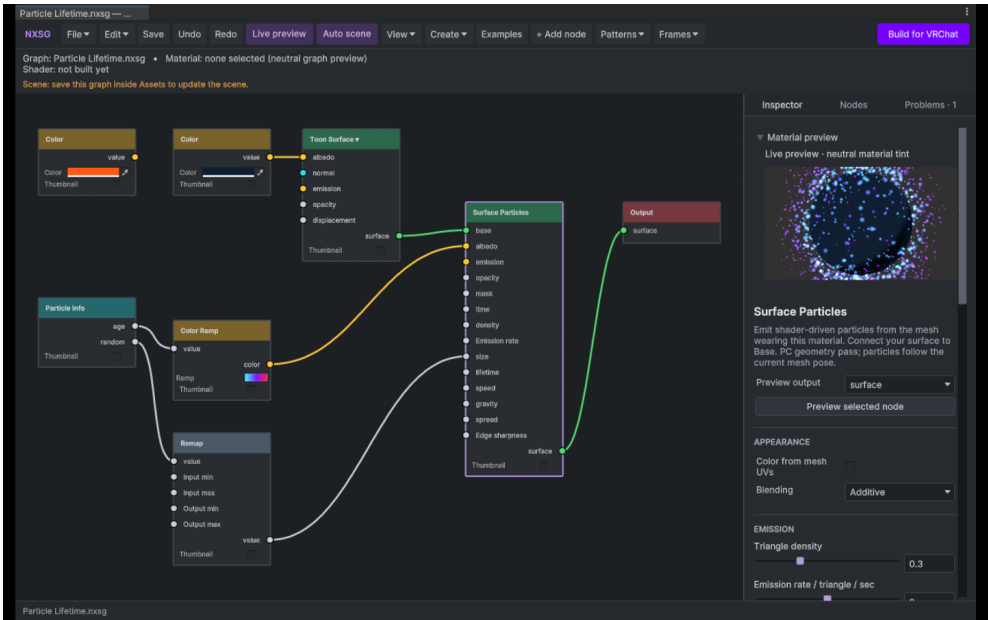
Canonical semantic hashes exclude layout and formatting. Moving a node therefore need not trigger a scene rebuild. Properties that affect compilation or resource meaning still change the semantic result. Mutable material and animation bindings cannot be optimized away merely because their defaults look neutral.

## FUTURE-PROOFING RULE

Specialize only proven constants. Preserve stable identities, unknown data and legacy defaults; migrate deliberately and report unsupported behavior rather than silently changing a graph.

# Authoring and the scene are connected

The editor now covers much more than dragging nodes.



Actual Unity editor capture from the repository UI pass. The image demonstrates layout and appearance, not all interaction states.

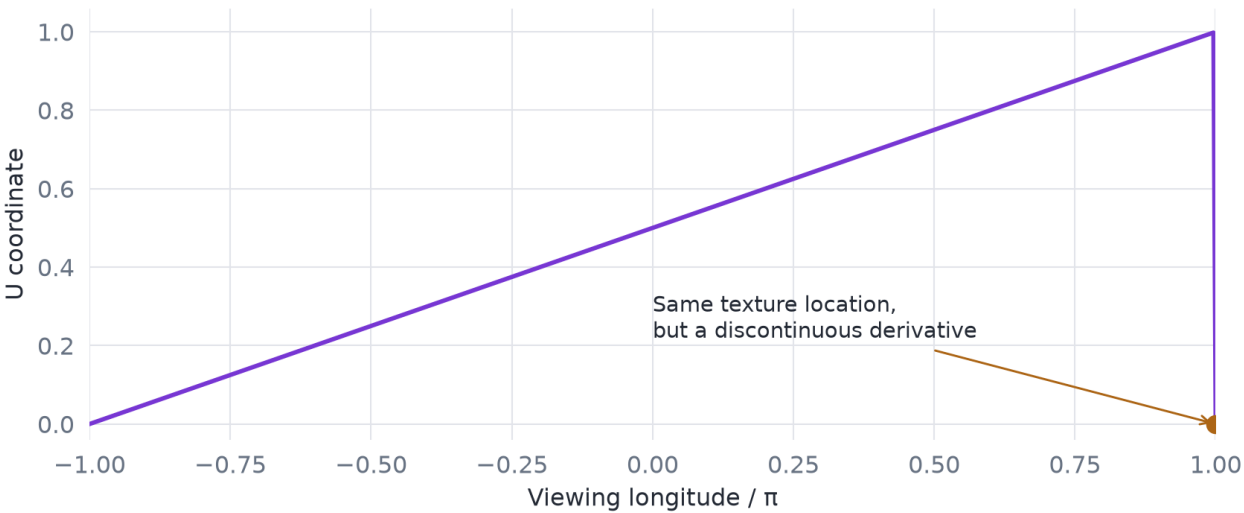
| Workflow             | Current tools                                                                                                                                        |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Build a graph        | Categorized search and descriptions; sockets in either direction; drop wire onto empty space to create a compatible node; related header operations. |
| Organize             | Selection outline, box select, copy/paste/duplicate, frames and notes, finder, bookmarks and editable copied Patterns.                               |
| Understand materials | Graph/material/shader identity, named textures, node output preview, Cost hotspots and a mismatch explanation tool.                                  |
| Compare / reuse      | Material playground, A/B snapshots, time scrubbing, lighting controls, presets and texture-set import.                                               |
| Recover              | Undo/Redo, last-valid preview, build diagnostics, output rollback and stable asset GUIDs.                                                            |

Patterns are independent copies, not linked external instances. Dense inspectors still scroll. The preview can collapse to make space, and the existing Linux keyboard-Undo limitation remains documented.

Sources: R01 editor; R12 UI polish; R13 creator workflow; R22 quick start.

# Coordinates and the Panosphere lesson

A seamless image is not enough when its sampling coordinates have a discontinuity.



**Fig. 11.** EXPLANATION. Longitude U wraps from 1 to 0. Adjacent pixels can be close on the texture while a naive derivative suggests a huge jump.

Mesh UV0-UV3, object/world projection, Polar, Panosphere and Matcap serve different purposes. Mesh and object coordinates follow the mesh or object; world coordinates are fixed in world space. Panosphere and Matcap intentionally depend on viewing direction. Camera-relative behavior is not interchangeable with an authored UV unwrap.

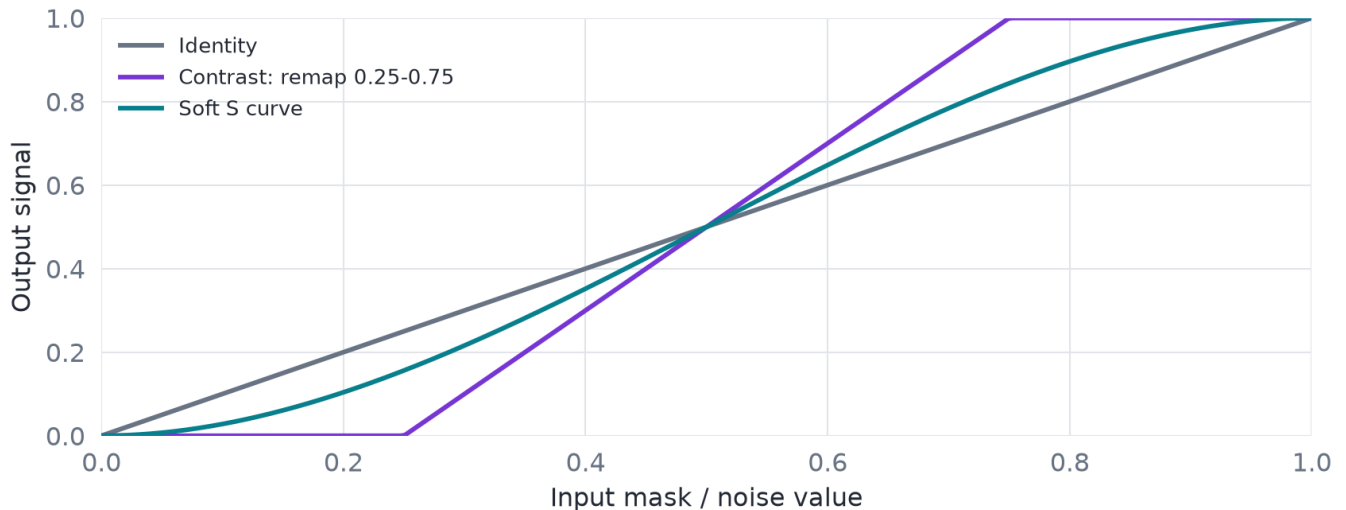
NXSG's Panosphere seam fix follows Poiyomi's derivative-aware approach, with attribution in the package notices. A later tolerance prevents numerically similar chart derivatives selecting different branches in neighboring pixels. The regression set also checks fractional tiling away from the wrap.

| Need                        | Use / caveat                                                                                    |
|-----------------------------|-------------------------------------------------------------------------------------------------|
| Stable painted layout       | Mesh UVs. UV seams and overlapping islands remain properties of the mesh.                       |
| Projected procedural detail | Object or world mapping. Choose the intended coordinate space deliberately.                     |
| Polar transformation        | Transforms an input coordinate field; its input-source selector does not replace the operation. |
| View-driven image           | Panosphere or Matcap. Recheck mirrors/stereo; no headset equivalence is claimed.                |

Sources: R14 PANOSPHERE; R05 nodes; R02 coordinate render checks; third-party notices.

# Procedurals, masks and numerical control

The graph can reshape a signal before it becomes color, opacity or motion.



**Fig. 12.** ILLUSTRATIVE SIGNAL CURVES. These are defined mathematical examples, not sampled output from a particular NXSG graph.

The procedural library includes 1D-4D noise, Voronoi, Musgrave-style fractals, checkerboards, waves and many pattern/effect generators. The chosen dimension determines whether the signal follows a scalar, UV plane, spatial position or a time-evolving field. Object and world space should be chosen intentionally.

Ramp controls black/white points and a curve on a numerical field, which is useful for sharpening noise into a mask. Color Ramp maps a signal into colors. Distortion and coordinate transforms can be composed before texture or procedural sampling. A seamless source does not guarantee that later warping or coordinate wrapping stays seamless.

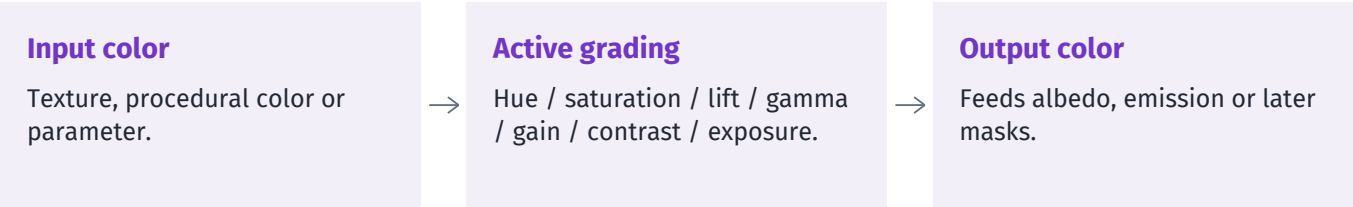
Math headers allow related operations to switch in place. Mix factors use a 0-1 semantic range. Wider numeric fields let authors enter values beyond convenient slider tracks where the operation permits them; finite-value validation and hard shader limits still apply.

## READ THE SIGNAL YOU ACTUALLY NEED

RGB brightness, texture alpha, a single channel and a geometric mask answer different questions. Use explicit Alpha or channel outputs when opacity should follow that data, rather than relying on a color conversion.

# Color adjustment and hue spaces

One node groups grading controls while keeping neutral constant stages out of generated work.



**Fig. 13.** IMPLEMENTED CONCEPT. Only constant stages proven neutral can disappear; connected controls remain dynamic.

| Control family   | Behavior to understand                                                                                                                     |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Hue space        | HSV and OKLab are selectable. HSV is the legacy default; old graphs preserve their interpretation.                                         |
| OKLab            | Operates in a perceptual color representation. Rotating hue while holding its lightness/chroma axes behaves differently from HSV rotation. |
| Neutral controls | The compiler omits eligible unchanged stages. This is specialization, not a guarantee of zero overhead for every connected graph.          |
| Color Mask       | Selects color by distance and softness to produce a useful mask.                                                                           |
| Replace Color    | Recolors selected regions with factor/alpha behavior and connected controls.                                                               |
| HDR / negatives  | The recorded hue tests include gray, HDR, negative channels and negative hue; alpha preservation is checked.                               |

The 1.3 render fixture compares both hue nodes with an independent OKLab reference in Gamma and Linear projects. That is stronger evidence than comparing two paths that share the same implementation. It still does not establish native Windows or headset rendering.

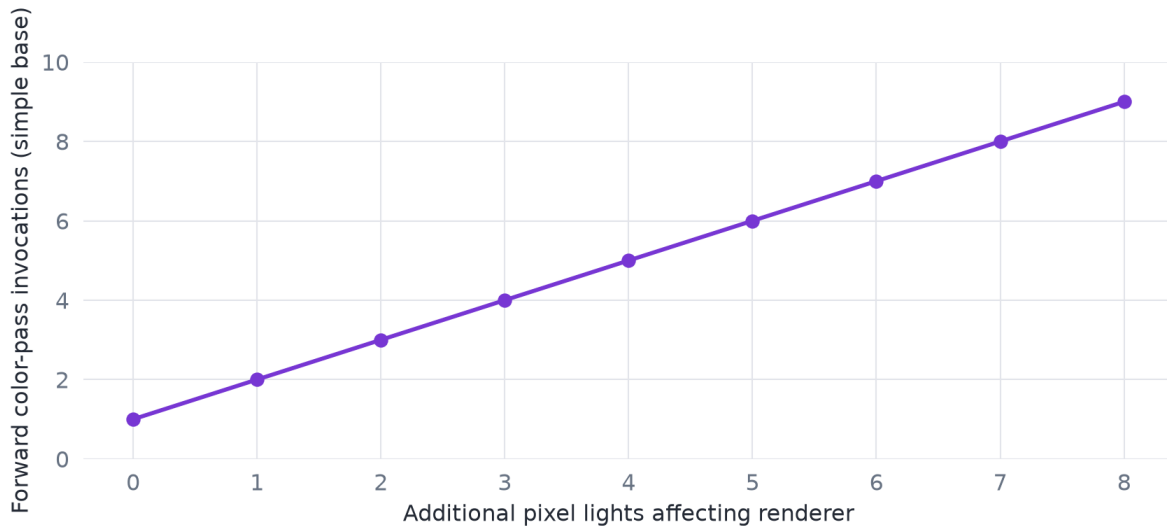
## ARTISTIC CONTROL, NOT AUTOMATIC COLOR MANAGEMENT

Hue space selection does not fix a wrongly imported texture or an unintended Gamma/Linear project mismatch. Keep source texture intent and project color space part of each repro.

Sources: R02 1.1-1.4 records; R05 nodes; work/v1.2-release-notes.md and v1.3-release-notes.md; W04 OKLab.

# Lighting, depth and self-shadowing

Surface orientation and occlusion are separate inputs to the lighting result.



**Fig. 14. EXPLANATORY COUNT.** A simple lit base needs one ForwardBase plus one ForwardAdd invocation per additional pixel light. Shadow, overlay and other cameras are excluded.

Toon and PBR base surfaces receive additional pixel lights through ForwardAdd. Ambient, reflections and emission remain in the base pass so they are not added repeatedly. Generated lighting uses Unity Built-In macros; selected checks do not cover every light/shadow variant. Shell/fur overlays retain a narrower lighting path.

Minimum/maximum brightness and lighting saturation tune incident lighting. A maximum of zero means unlimited. Limits apply per contribution; the sum of several lights can still exceed a single contribution's limit. These controls affect lighting, not albedo or emission.

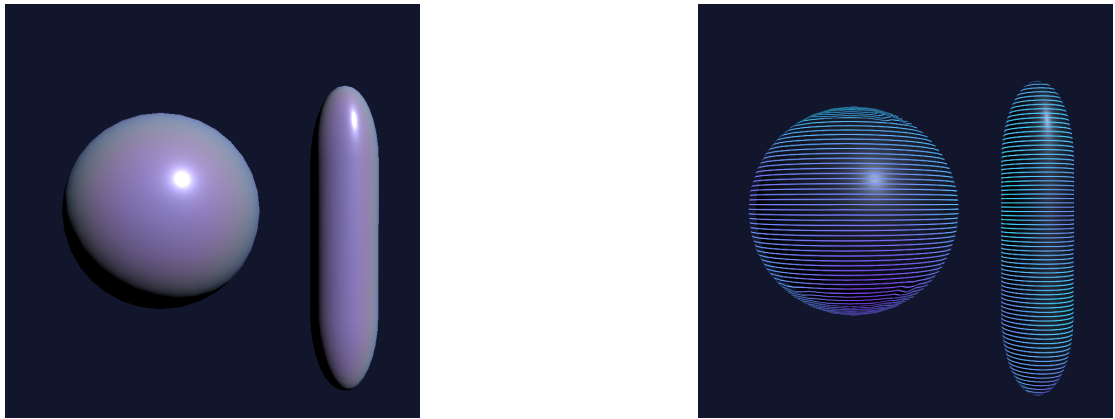
A normal facing a light can receive direct lighting even if another part of the avatar lies between it and the light. The material needs an occlusion signal to know that. Scene shadow maps can provide one; a local fur-volume approximation cannot create exact cross-body shadows across separate material regions.

## OPAQUE COVERAGE STILL NEEDS CORRECT DEPTH

Opacity 1 with albedo alpha ignored should retain opaque coverage. The fixed dependency bug shows why ShadowCaster/depth behavior must be checked independently from the visible alpha value.

# Surfaces and material layers

Toon, PBR and Unlit form the base; other nodes add deliberate behaviors.



Actual Unity material studies from the repository: pearl/iridescence and layered hologram. Presentation setup is distinct from headset performance evidence.

| Surface path       | What it adds / limits                                                                                                                                             |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Toon / PBR / Unlit | Stylized lighting, metallic/roughness lighting, or lighting-independent color. Normal, opacity, emission and displacement depend on the selected surface.         |
| Layered PBR        | Clearcoat weight, independent roughness and normal; colored velvet sheen with weight and roughness. Artistic approximations, not a full glTF BSDF implementation. |
| Shell              | Offset another surface layer; nested shell stacks are supported. Transparency, bounds and pass count need attention.                                              |
| Particle Surface   | Material for Unity's existing ParticleSystemRenderer, distinct from same-mesh Surface Particles.                                                                  |
| Volume Surface     | Ray integration or solid SDF in a closed proxy. It has its own output/composition constraints.                                                                    |

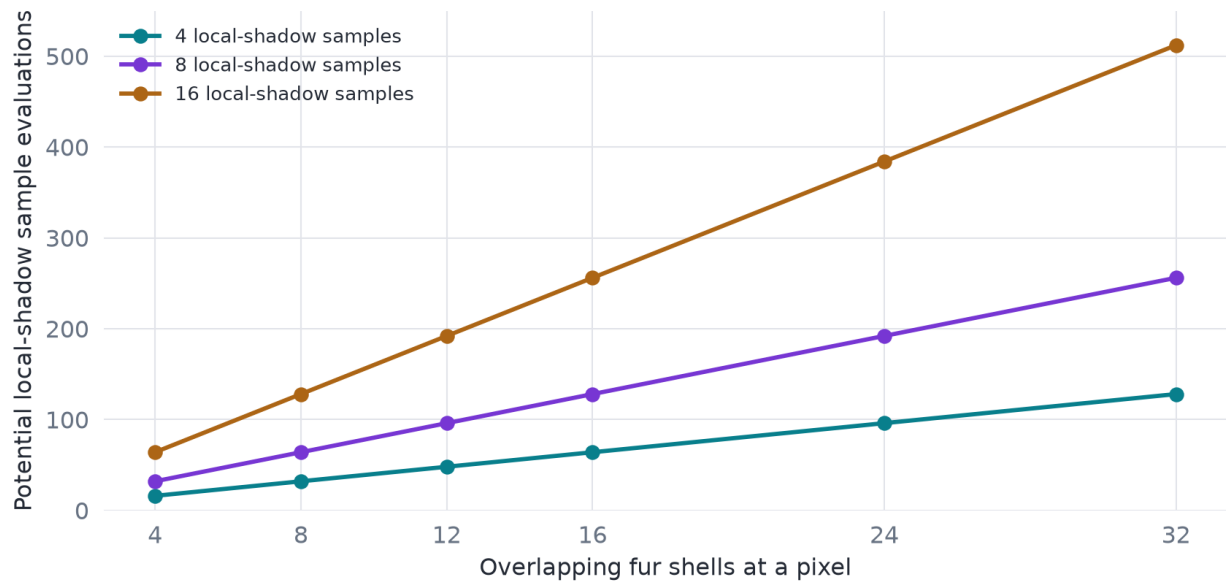
Use albedo alpha is an explicit coverage choice. With it disabled, an albedo texture should not silently make the base transparent. Connected opacity, cutoff, blend state and overlay layers still determine their respective coverage.

Glitter supplies view-dependent speckles as color or a numerical mask. Wireframe reveals triangle edges. Sticker places a bounded texture patch; its placement tool edits UV controls rather than painting pixels. Refraction samples the screen, while Interior Mapping suggests depth without creating an actual room mesh.

Sources: R01 features; R02 layered surface tests; R05 node catalog; source graphs and retained Unity renders.

# Fur: shells, fins and generated cards

Three geometry choices with different silhouettes and work.



**Fig. 15.** MODEL. Maximum local-shadow sample work at a pixel is approximated here as overlapping shells x samples per shell fragment. No GPU milliseconds are implied.

| Mode          | Current behavior                                                                                                                                             |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shells        | Layered offset surfaces. The Groomed Fur sample uses 16 layers; a showcase uses 24. Layer counts are appearance settings, not validated performance budgets. |
| Fins          | One geometry pass creates three edge strips per source triangle. It improves grazing silhouettes without a mesh-adjacency solution.                          |
| Cards only    | Three cards from selected triangle edges, with no shells. Shared edges can overlap. Generated in the shader, not exported as a card mesh.                    |
| Local shadows | Main-light strand-volume approximation: off, 4, 8 or 16 samples. It does not exactly trace bent strands or cross-body occlusion.                             |

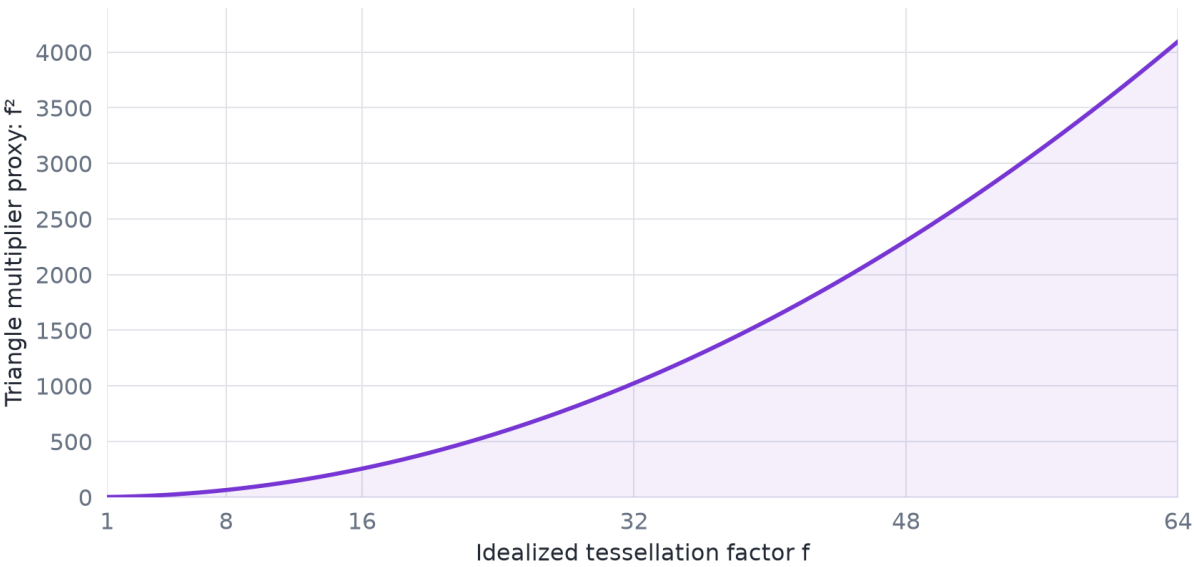
Root/tip colors, masks, grooming and wind remain artist controls. Fur receives main directional shadows; its base casts the scene shadow. Additional pixel lights do not light the overlays. Expand renderer bounds for long fur. Distance LOD can reduce active coverage without erasing all submitted shell draws.

The useful future experiment is a matched shells/fins/cards comparison on the same skinned mesh, camera path and target GPU, with silhouette quality and shadow cost recorded together.

Sources: R25 FUR\_AND\_PARALLAX; Core/GraphPerformance.cs; R17 rendering research.

# Parallax and tessellation

One changes sampling; the other changes the geometry sent down the pipeline.



**Fig. 16.** MODEL.  $f^2$  is an intuitive triangle-growth proxy, not the exact backend topology count. Partitioning and adaptive factors matter. No timings were measured for this plot.

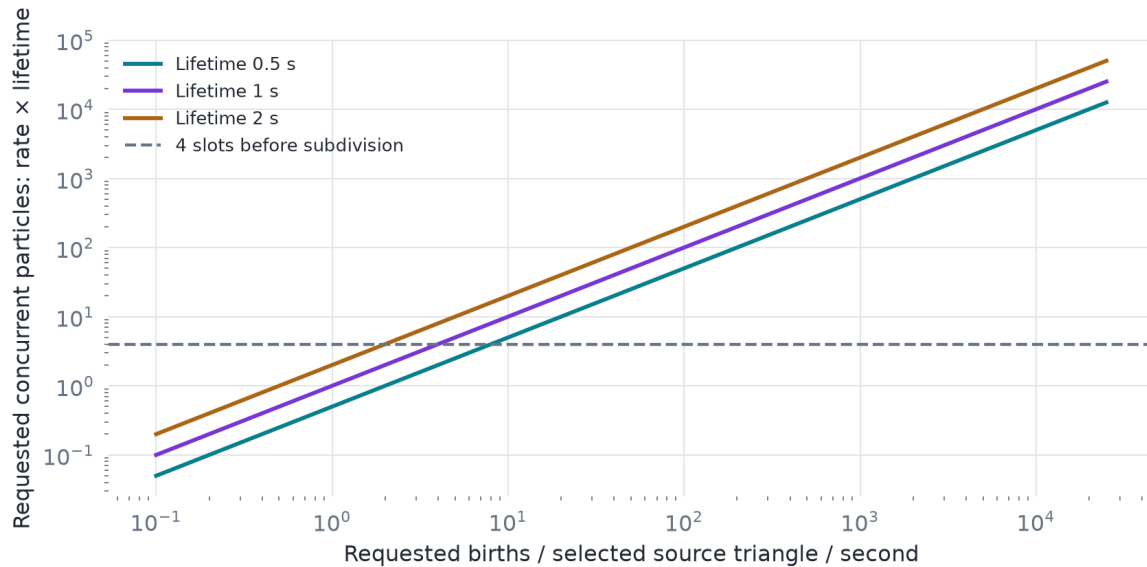
| Technique          | Current contract                                                                                                                                    |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Parallax occlusion | Repeated height sampling shifts texture coordinates. Needs tangents. Does not change the physical silhouette or cast displaced geometry shadows.    |
| Tessellation       | Subdivision precedes displacement. Factor range 1-63 with distance controls and optional smoothing; target 4.6 hardware support required.           |
| Displacement       | (height - reference height) x strength along the interpolated normal. Existing displacement is additive; normal detail is authored separately.      |
| Composition        | One Tessellation before Output; Toon, Unlit, PBR or Shell stack supported. Fur and Surface Particles cannot be combined with that decorator.        |
| Seams / bounds     | Shared edge factors cannot fix discontinuous UV height or split normals. Mesh assets, colliders and renderer bounds are not expanded automatically. |

A high factor on an already dense avatar can create much more work than on a low-poly test sphere. Evaluate the actual source triangle count and screen coverage. A mask that hides final pixels does not necessarily remove upstream subdivision or geometry processing.

Sources: R25 parallax; R26 TESSELLATION; Core/GraphPerformance.cs; report equation is a simplified model.

# Particles emitted from the existing mesh

The shipped solution follows the current pose and computes motion analytically.



**Fig. 17. MODEL OF REQUESTED OCCUPANCY.** rate x lifetime is the desired concurrent population per selected source triangle. Actual capacity is bounded and tessellation-based.

Surface Particles sits between a base surface and Output. It keeps the base and emits camera-facing quads from source triangles without an extra mesh or Particle System. Four slots run per generated subtriangle. Requested occupancy above four enables subdivision; factor is capped at 64, and extreme requests plateau.

Density selects a fraction of triangles, not uniform surface area. Masking and zero density hide results but do not remove every geometry invocation. Source UV color sampling is optional; it samples connected Albedo/Emission at the spawn UV, while sprite opacity keeps sprite UVs. Vertex colors only apply when explicitly wired.

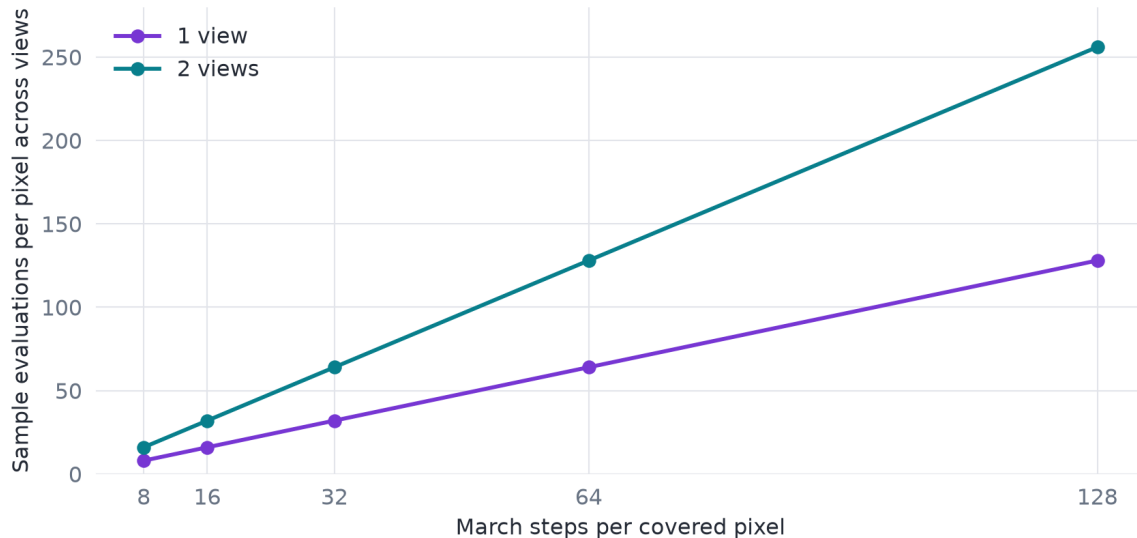
Numeric sockets expose density, rate, size, lifetime, speed, gravity, spread, sharpness, opacity, mask and time. Curves control size/opacity/color over age. Particle Info supplies age and a repeating slot-random value in permitted particle branches.

## WHAT “SHADER PARTICLES” MEANS HERE

No stored world-space birth history, collisions or persistent trails. Rate/lifetime changes retime the analytical loop. Skinning follows the current pose, base displacement is not inherited, bounds must cover motion, and alpha particles are not individually sorted.

# Raymarching and signed distance fields

A separate experimental output path, with a cost repeated over pixels and views.



**Fig. 18.** UPPER-BOUND MODEL. Covered pixels x executed steps x views drives field evaluation work. Early termination and proxy coverage change actual cost.

Volume Surface integrates density, color and emission inside an object-space box. Ray Position exposes the current march point; Sphere, Box, Torus and Blend define signed-distance fields. Density noise can use 3D or time-evolving 4D coordinates.

The step range is 8-128, default 32. Integration exits when accumulated opacity reaches 0.995. Solid SDF instead finds a hit and evaluates six distance samples for a normal. Its simple light model has no shadow reception/casting, extra lights or depth-writing pass.

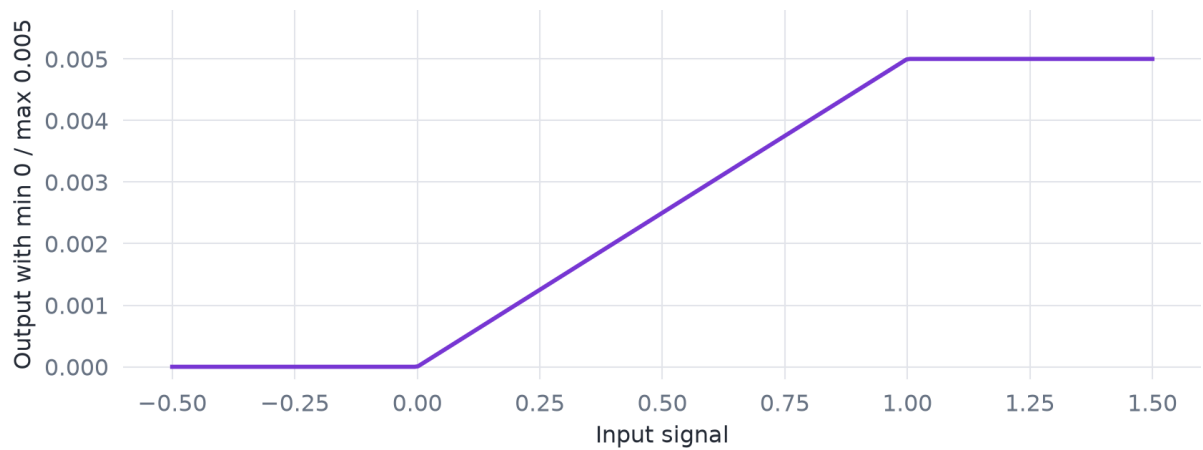
Use a closed box proxy with matching extents. An arbitrary avatar mesh does not become a correct volume proxy. Shell, Fur, Tessellation and Surface Particles cannot share this output path. Transparent sorting, scene-depth availability and texture sampling choices remain important limitations.

## RENDERED EXAMPLES ARE APPEARANCE EVIDENCE

The package includes six volume examples. Studio studies use presentation bloom/toning; basic studies are direct Unity captures. None establishes live headset frame time or mirror acceptance.

# Audio, motion and external lighting

Three integrations with three different sources of input data.



**Fig. 19.** ILLUSTRATIVE RANGE MAPPING. A clamped input mapped to min 0 / max 0.005 stays in that range. Gain alone is a different operation.

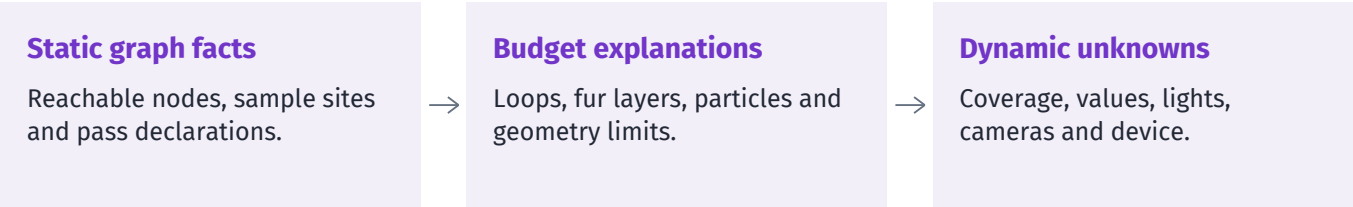
| Integration   | What is implemented                                                                                                                               | Boundary                                                                                                                                    |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| AudioLink     | Texture-backed band reading, smoothing, fallback and preview values; range mapping supports explicit endpoints.                                   | Synthetic 128x64 provider tests are recorded. Preview slider is one value, not live spectrum. Live provider/client validation remains open. |
| LTCGI         | Optional receiver node, albedo/normal/roughness inputs, additive emission contribution and receiver tag.                                          | Needs installed LTCGI and world emitters. Does not make the avatar an emitter. Fragment-only; omitted from extra-light and shadow paths.    |
| Avatar Motion | Material properties driven by a generated Animator controller or merged FX layers; speed/directional velocity into response, sway and UV stretch. | Requires driver setup. Measures player locomotion, not hands, bones or PhysBones; no physical inertia or stored motion history.             |

Use stable exposed properties for animation. The generated driver uses Unity blend trees and clips rather than a custom runtime component. VRCFury is an optional workflow around ordinary material properties, not a mandatory private API dependency.

Sources: R29 AudioLink implementation; R30 LTCGI; R31 motion/handles; R13 playground; R02 alpha.24.

# Cost estimates: useful, not a profiler

The tool can explain generated work while leaving dynamic GPU time unknown.



**Fig. 20.** COST MODEL. The panel can focus costly nodes; it does not turn these facts into measured milliseconds.

| Reported item        | Interpret it carefully                                                                                                           |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Texture sample site  | A location in generated evaluation. May execute in multiple stages/passes or inside a loop.                                      |
| Static passes        | Declared pass budget. ForwardAdd may run per extra pixel light; variants are not draw counts.                                    |
| Parallax loop        | An initial height read plus up to the configured iterations; early exit changes actual reads.                                    |
| Volume march         | Up to the configured steps per covered pixel per eye; each field can itself be expensive.                                        |
| Fur shells           | Shell passes and local-shadow sample choices multiply work. Discarding far fragments is not the same as removing all draw setup. |
| Particle subdivision | Connected rate/lifetime can make the geometry budget dynamic. Source triangle count multiplies the generated work.               |

The 1.5 checks compare static pass estimates to emitted declarations for all shipped samples. That is a consistency test of the estimator. It does not compare predictions to measured GPU costs, and no such timing model is advertised in this report.

### PRACTICAL USE

Use the panel to identify a candidate branch to simplify. Then compare appearance and actual runtime on a controlled scene before assigning a performance tier.

Sources: Core/GraphPerformance.cs; R02 1.5 Cost tests; R17 performance research.

# Keep graphs portable and inputs bounded

Data interchange and executable extensions must remain different trust boundaries.

| Mechanism                      | Current or documented behavior                                                                                                 |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Canonical serialization        | Stable IDs and deterministic JSON. Layout is separate from semantic meaning; unknown fields are retained where supported.      |
| Malformed input                | Duplicate keys/IDs, invalid references, bad ports, cycles, nonfinite numbers and resource traversal are rejected or diagnosed. |
| Parser / authored graph limits | Documented defaults: 4 MiB input, depth 64, 4,096 nodes, 16,384 connections, 1,024 resources and 32 nested Pattern levels.     |
| Clipboard                      | A bounded snippet with identity remapping. The historical implementation record specifies 1 MiB and 1,024 nodes per snippet.   |
| Unknown operations             | Inert preserved data; cannot silently execute downloaded code or masquerade as a supported operation.                          |
| Future packs / bridges         | Raw HLSL/C# is executable trust. Blender/MaterialX/glTF translation must say exact, approximate, baked or unsupported.         |

The original graph contract is an early slice. Its list of only seven initial operations is historical; the current catalog is much larger. Similarly, reserved future backend limits in that document should not be presented as enforced just because they are written down.

Static baking is limited to supported UV-local numerical/color branches. It writes linear, non-HDR 8-bit PNG at selected sizes, clamps negative/HDR values and keeps the source graph editable. View, time, pose and other unsupported dependencies require a live path or an explicit approximation.

## PRESERVE USER INTENT

Keep migrations explicit, defaults compatible, asset metadata stable and save failures recoverable. A future adapter should describe losses, never silently claim identical rendering.

Sources: R21 graph contract; R09 compiler research; R08 decisions; R20 validation/release research; current validation.

# What the audit found beyond code

A mature implementation needs documentation that agrees with its own chronology.

| Finding                           | Why it matters                                                                | Action / status                                                                                      |
|-----------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Stale node totals                 | Current Add menu exposes 155; feature count is 52 rather than 45.             | Fixed locally, including the 52-name feature list. Future improvement: generate these automatically. |
| Historical pending claims         | Early contract/research sections describe the smaller initial implementation. | Retained as dated history. Read the current README and later validation entries for present scope.   |
| Old validation header             | Opening status was Sep 19 while evidence reaches Sep 24.                      | Updated locally to Sep 24; dated historical entries preserved.                                       |
| Ambiguous D3D wording             | “Windows/D3D unverified” could imply no D3D compilation exists.               | Corrected locally: 16-shader D3D11 cross-compilation passed; native rendering unverified.            |
| License remains undecided         | Public source does not automatically grant broad reuse rights.                | Owner decision needed; no license selected by this report.                                           |
| No broad runtime performance data | Local functionality evidence is not a headset budget.                         | Future work: reproducible hardware/client performance records before preset claims.                  |

These findings do not invalidate the released feature set. They identify places where a reader can get conflicting answers depending on which document they open. A generated catalog and a short current-status summary would remove much of that ambiguity.

## CORRECTIONS MADE WITH THIS REPORT

NODES.md, VALIDATION.md and COMPATIBILITY.md were corrected in the local checkout. The feature-name list was checked against the catalog and `git diff --check` passed. Shader code, user graphs/materials and publication state were not changed.

Sources: R01 README; R02 validation; R03 catalog audit; R05 node guide; R16 research index; R21 graph contract.

# Node catalog / inputs and patterns

All Add-menu entries, grouped exactly as the current editor categorizes them.

## INPUTS / 18

Audio Link  
Avatar Motion  
Camera Distance  
Color  
Distance to Point  
Normal Direction  
Object Origin  
Object Random  
Object Scale  
Parameter  
Particle Color  
Particle Info  
Position  
Reflection Direction  
Time  
Value  
Vertex Color  
View Direction

## COORDINATES / 22

Barrel Distortion  
Combine UV  
Flow Map UVs  
Glitch UVs  
Kaleidoscope UVs  
Motion Stretch UVs  
Object Planar UVs  
Parallax Occlusion  
Parallax UVs  
Pinch UVs  
Pixelate UVs  
Polar UVs  
Rotate UVs  
Screen UVs  
Spherize UVs  
Split UV  
Swap UV Axes

## COORDINATES / CONTINUED

UV Coordinates  
UV Scroll  
UV Tile / Mirror  
UV Transform  
World Planar UVs

## TEXTURES / 30

Box Volume Mask  
Capsule Volume Mask  
Checkerboard  
Chromatic Texture  
Clouds  
Cracks  
Dither Mask  
Fur Strand Mask  
Glitter  
Gradient  
Hologram Scanlines  
Marble  
Matcap Texture  
Musgrave  
Noise  
Polka Dots  
Scale Pattern  
Scratches  
Snow Coverage  
Sparkle Mask  
Sphere Volume Mask  
Texture  
Texture Bomb  
Triplanar Texture  
Truchet Tiles  
Volume Stripes  
Voronoi  
Waves  
Wood Rings  
Woven Fabric

Sources: R03 code-derived inventory at 4e36698; source/inventory.json. 155 visible nodes across these three pages.

# Node catalog / values and color

All Add-menu entries, grouped exactly as the current editor categorizes them.

## MATH / 31

- Absolute
- Add
- Box Mask
- Brick Pattern
- Circle Mask
- Clamp
- Cosine
- Divide
- Fraction
- Hex Grid
- Invert
- Maximum
- Minimum
- Mix
- Multiply
- Ping Pong
- Polygon Mask
- Power
- Radial Rays
- Ramp
- Remap
- Round
- Round Down
- Round Up
- Sine
- Smoothstep
- Spiral
- Square Root
- Star Mask
- Step
- Subtract

## COLOR / 25

- Color Adjust
- Color Mask
- Color Ramp
- Combine Color
- Contrast
- Dissolve
- Distance Fade
- Fresnel
- Height Mask
- Hue Shift
- Interior Mapping
- Iridescence
- Layer
- Luminance
- Posterize
- Refraction
- Replace Color
- Rim Glow
- Saturation
- Slope Mask
- Split Color
- Sticker
- Subsurface
- Wet Color
- Wireframe

## ANIMATION / 5

- Distortion
- Flipbook
- Motion Response
- Motion Sway
- Vertex Motion

# Node catalog / rendering outputs

All Add-menu entries, grouped exactly as the current editor categorizes them.

## **SURFACE / 18**

Anisotropic Highlight  
Blend Normals  
Darkness Glow  
Emission  
Fur  
Layered PBR Surface  
LTCGI Lighting  
Normal from Height  
Normal Map  
Normal Strength  
Output  
Particle Surface

## **SURFACE / CONTINUED**

PBR Surface  
Shell  
Surface Particles  
Tessellation  
Toon Surface  
Unlit Surface  
**VOLUMES / 6**  
Ray Position  
SDF Blend  
SDF Box  
SDF Sphere  
SDF Torus  
Volume Surface

Display names are not stable IDs. The accompanying inventory records each operation ID, name and category. Node behavior also depends on its dropdown modes, so an operation count is not a count of every available effect or combination.

### **INTERNAL HELPER**

The registered `core.previewVector` operation is excluded from these lists because the editor hides it from the Add menu. Float/Color parameter workflows can be special editor actions around registered parameter support.

Composition limits still apply: for example, not every Surface output can be nested, and Volume Surface is a separate output path. The complete list is a discovery reference, not an assurance that every arbitrary connection is valid.

# Every bundled example graph

Counts describe authored topology, not runtime complexity or test coverage.

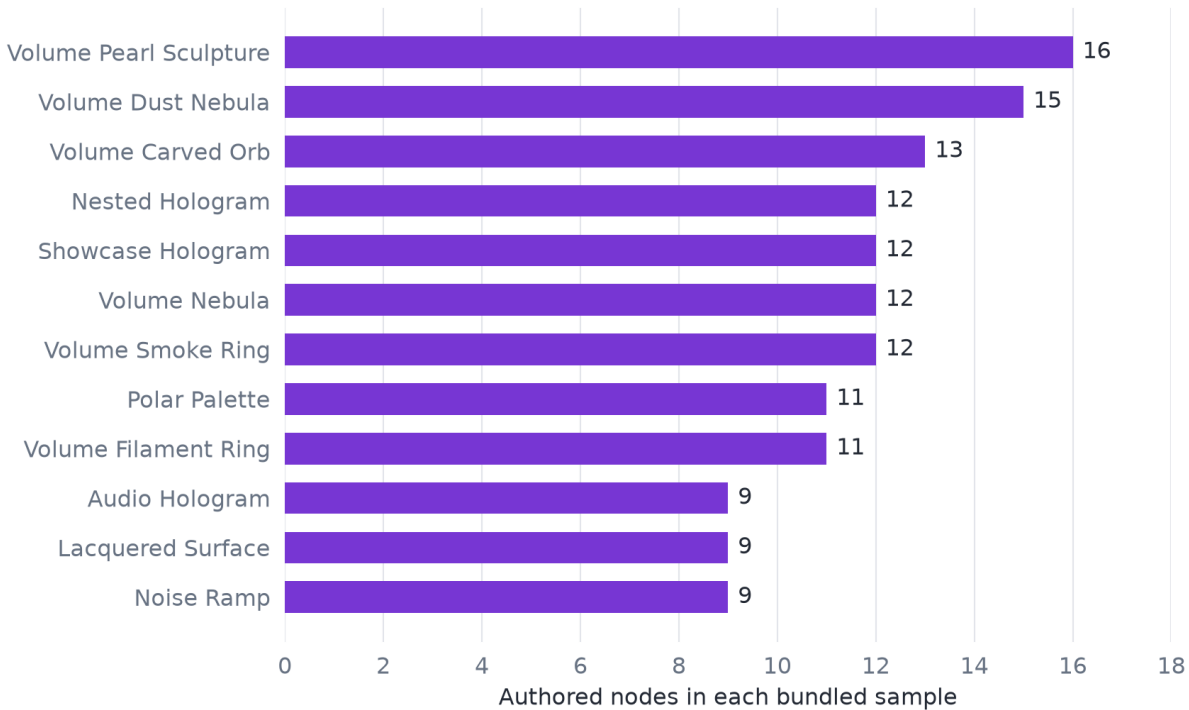
| Bundled graph     | N  | E  | Bundled graph          | N  | E  |
|-------------------|----|----|------------------------|----|----|
| 4D Clouds         | 5  | 4  | Particle Sparkles      | 7  | 6  |
| Animated Palette  | 8  | 8  | Polar Palette          | 11 | 11 |
| Animated Sticker  | 7  | 6  | Refraction Glass       | 5  | 4  |
| Audio Hologram    | 9  | 11 | Ripple Tiles           | 9  | 9  |
| Fur Cards         | 7  | 6  | Shiny Surface          | 6  | 6  |
| Fur Fins          | 7  | 6  | Showcase Hologram      | 12 | 13 |
| Glitter Fabric    | 6  | 5  | Showcase Pearl         | 7  | 8  |
| Groomed Fur       | 7  | 6  | Showcase Warm Fur      | 8  | 7  |
| Interior Bomb     | 6  | 6  | Surface Sparkles       | 5  | 4  |
| Lacquered Surface | 9  | 8  | Tessellated Bumps      | 6  | 5  |
| LTCGI Receiver    | 4  | 4  | Velvet Fabric          | 9  | 8  |
| Motion Glow       | 8  | 8  | Volume Carved Orb      | 13 | 13 |
| Neon Wireframe    | 6  | 5  | Volume Dust Nebula     | 15 | 18 |
| Nested Hologram   | 12 | 14 | Volume Filament Ring   | 11 | 13 |
| Noise Color Ramp  | 6  | 5  | Volume Nebula          | 12 | 15 |
| Noise Ramp        | 9  | 9  | Volume Pearl Sculpture | 16 | 20 |
| Parallax Tiles    | 5  | 4  | Volume Smoke Ring      | 12 | 15 |
| Particle Lifetime | 8  | 7  |                        |    |    |

N = authored nodes. E = explicit connections (wires). Every file is included once; neither layout-only helpers nor visual complexity are added to these counts.

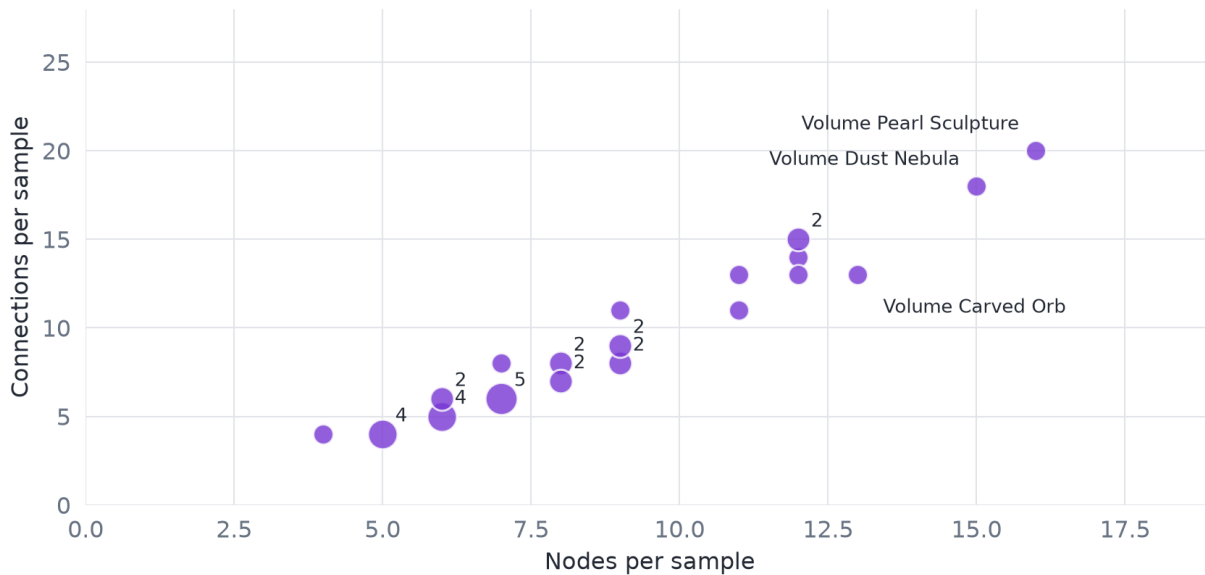
Sources: R03 inventory: JSON parse of all Samples-/\*.nxsg. 35 files, 293 nodes, 297 connections.

# How the sample library is shaped

Examples favor small, inspectable graphs; a larger graph is not necessarily more expensive.



**Fig. 21. INVENTORY.** The 12 largest bundled graphs by authored node count. No cost weighting is applied.



**Fig. 22. INVENTORY.** Node/wire counts of the 35 examples; numbers on coincident points show how many graphs share that topology count.

Sources: R03 inventory; package Samples~ graph JSON. No rendering or test coverage inferred from these counts.

# Repository sources and how to trace them

References point to the audited revision, not an unversioned future README.

**R01** [README.md](#)

Product scope, stable release, compatibility and license status.

**R02** [docs/VALIDATION.md](#)

Dated records; latest 1.5 at baseline lines 566-580. Header corrected locally.

**R03** [Packages/dev.nerdrx.nxsg/Core/NodeCatalog.cs](#)

Catalog plus FeatureNodes.cs, GraphWindow.cs filter and Samples~ JSON.

**R04** [docs/COMPATIBILITY.md](#)

Platform boundaries; D3D wording corrected locally after the audit.

**R05** [docs/NODES.md](#)

Node guide; counts and feature list corrected locally after the audit.

**R06** [docs/RISK\\_REGISTER.md](#)

Risk owners, mitigations and evidence gates.

**R07** [docs/IMPLEMENTATION\\_PLAN.md](#)

Original milestone dependencies and acceptance criteria.

**R08** [docs/DECISIONS.md](#)

Selected design directions and deferred alternatives.

**R09** [docs/research/GRAPH\\_COMPILER\\_AND\\_PORTABILITY.md](#)

Types, graph format, compiler and interchange research.

**R10** [work/unity/v1.5-\\*.log](#)

Local retained test logs; exact markers on page 39.

**R11** [docs/BUILD\\_PERFORMANCE.md](#)

Three warm build observations and interpretation limits.

**R12** [docs/UI\\_POLISH.md](#)

Actual editor styling, bounds and layout checks.

**R13** [docs/CREATOR\\_WORKFLOW.md](#)

Playground, presets, baking and texture import.

**R14** [docs/PANOSPHERE.md](#)

Sampling seams, derivative chart selection and attribution.

**R15** [docs/VPM.md](#)

Package workflow; scripts/build-vpm.py and verify-vpm.py.

**R16** [docs/RESEARCH\\_INDEX.md](#)

Research map and original evidence boundary.

Sources: All paths below are relative to nerdrx/nxsg at 4e36698 unless described as local logs.

---

# Research and primary references

Selected primary sources were opened on 25 September 2026. Older research citations retain their own recorded dates.

**R17** [docs/research/RENDERING\\_FUR\\_AND\\_PERFORMANCE.md](#)

Fur, rendering and measurement.

**R18** [docs/research/PLATFORM\\_AND\\_INTEGRATIONS.md](#)

Platform, policy and integration research.

**R19** [docs/research/EDITOR\\_UX\\_AND\\_PRIOR\\_ART.md](#)

UX and prior-art research; custom-canvas report is adjacent.

**R20** [docs/research/VALIDATION\\_AND\\_RELEASE.md](#)

Testing, trust, licensing and release.

**R21** [docs/GRAPH\\_CONTRACT.md](#)

Early portable contract; treat its small catalog as historical.

**R22** [docs/QUICK\\_START.md](#)

Editor and build workflow.

**R23 / R24** [docs/PIXEL\\_LIGHTS.md](#)

Lighting behavior; lighting tests are in R02.

**R25 / R26** [docs/FUR\\_AND\\_PARALLAX.md](#)

Fur/parallax; tessellation is in docs/TESELLATION.md.

**R27 / R28** [docs/PARTICLES.md](#)

Particles; volumes are in docs/VOLUMES.md.

**R29 / R30 / R31** [docs/research/AUDIOLINK\\_IMPLEMENTATION.md](#)

Also docs/LTCGI.md and docs/MOTION\_AND\_HANDLES.md.

**R32** [docs/research/GPU\\_PARTICLES.md](#)

Particle feasibility research and alternatives.

**R33** [docs/research/CUSTOM\\_CANVAS\\_ON\\_UNITY\\_2022.md](#)

Pinned-version canvas/GraphView comparison.

## Primary references reopened for this report

**W01** [VRChat: supported Unity version](#)

**W02** [Unity 2022.3: camera depth textures](#)

**W03** [Creator Companion: VPM packages](#)

**W04** [Bjorn Ottosson: the original OKLab description](#)

**W05** [Unity 2022.3: forward rendering](#)

**W06** [VRChat: Android content limits](#)

**W07** [LTCGI: official documentation](#)

Sources: Primary-source links below; remaining research chronology is in R16 and the repository decision log.

---

# Evidence manifest and reproducibility

The report separates a new source audit from historical executable test results.

Source and package manifests were audited at revision 4e36698. Three documentation files were then corrected locally, as listed on page 31. Pre-existing Unity fixture settings and user test assets were left untouched. No new full Unity, headset or client regression suite was run for this PDF. The live listing and archive checksum were rechecked.

| Retained log / physical line    | Observed marker                                                                                                                               |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| v1.5-layered3.log:441           | LAYERED SURFACE SMOKE PASSED: neutral PBR, connected zero, coat roughness/normal, sheen roughness, black/grazing finite, emission, ForwardAdd |
| v1.5-layered-ui.log:499         | LAYERED CONTROLS PASSED: header switching, dormant properties, scalar/color edits, connected fields, bounds and socket labels                 |
| v1.5-cost-final.log:448         | PERFORMANCE SMOKE PASSED: estimates render and hot spot focus opens Inspector                                                                 |
| v1.5-thumbnail-lifetime.log:386 | THUMBNAIL LIFETIME PASSED: repeated paused deletion releases thumbnail editors, materials and shaders                                         |
| v1.5-build-transaction.log:542  | BUILD TRANSACTION PASSED: unchanged, changed, rollback, GUIDs, corrupt output, missing material, compiler failure                             |
| v1.5-scene-sync.log:513         | SCENE SYNC SMOKE PASSED                                                                                                                       |
| v1.5-d3d.log:1677               | D3D COMPILE SMOKE PASSED: 16 shaders                                                                                                          |

The portable log also records its success marker at work/v1.5-portable-final.log:2. Eleven packaging tests are reported by the validation document; this audit did not independently reconstruct each historic packaging-test result. The 1.4 timing is document-backed, with no matching original timing log found.

Catalog counts reproduce the Add-menu filter, not a marketing total. Sample counts parse all .nxsg files in Samples~. Measured charts use the recorded timing table. Cost curves are explicit algebraic models. They are never presented as GPU measurements or statistical distributions.

## WHAT THIS REPORT LEAVES YOU WITH

A current feature map, a realistic next-step program, traced evidence, complete discovery catalogs and specific documentation fixes. The next development cycle can now target the weakest evidence instead of adding features blindly.

Editable report source and machine-readable data are retained locally with this PDF. Figures use original calculations or the project's existing Unity captures; no synthetic showcase images are substituted for tested output.

Sources: source/evidence.json, source/inventory.json and source/live-vpm-listing.json beside the report builder.